

CO2 - RT2 - SCENARIO BASED QUESTIONS

🔍 DOKU

QUESTIONS

Q1

Scenario Based - Smart Waste Recycling Plant System
25 marks

Q2

Scenario Based - IT Company Project Performance Evaluation (2025-2026).
25 marks

Q3

Scenario Based - 3. Semiconductor Chip Manufacturing System
25 marks

Q4

Scenario Based - 4. Cloud Storage Subscription Management (2026)
25 marks

4/4 answered

Q3 Scenario Based - 3. Semiconductor Chip Manufacturing System

25 marks

A semiconductor company manufactures 10,000 chips per day.

Quality Standards:

Defect Rate < 1% → Grade A

1-3% → Grade B

3-5% → Grade C

Above 5% → Reject Batch

A) Develop a C program using arithmetic operators to calculate the defect percentage and accepted chips. Use decision-making statements to assign quality grades.

B) Use looping constructs to process production for 30 days. Use continue for maintenance days and break if a machine breakdown occurs.

Your Code

```
1 #include<stdio.h>
2 int main(){
3     int total=10000,defective,accepted,m,f;
4     float rate;
5     //part a
6     scanf("%d",&defective);
7     rate=((float)defective/total)*100;
8     accepted=total-defective;
9     printf("defect rate: %.2f%%\n accepted chips: %d\n",rate,accepted);
10    if(rate<1.0)printf("grade A \n");
11    else if(rate<=3.0)printf("grade B \n");
12    else if(rate <=5.0)printf("grade C \n");
13    else printf("reject batch \n");
14    //part b
15    for(int i=1;i<=30;i++){
16        scanf("%d",&m);
17        if(m==1)continue;
18        scanf("%d",&f);
19        if(f==1)break;
20        scanf("%d",&defective);
21        rate=((float)defective/total)*100;
22        printf("dat %d defect rate : %.2f%%\n",i,rate);
23    }
24    return 0;
```

Input

50
0 0 150
0 0 250
0 1

Output

Activate Windows
Go to Settings to activate Windows.

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Q2 Scenario Based - IT Company Project Performance Evaluation (2025–2026).

25 marks

An IT company completed 20 software projects during the financial year 2025–2026. Each project is evaluated based on Quality Score (100 marks) and Delivery Delay (days).

Performance Criteria:

Quality ≥ 90 and Delay ≤ 2 days → Excellent

Quality 75–89 and Delay ≤ 5 days → Good

Quality 60–74 → Average

Below 60 → Needs Improvement

A) Develop a C program using arithmetic operators to calculate the average quality score and delivery percentage. Use decision-making statements to classify each project.

B) Use looping constructs to process all 20 projects. Use continue to ignore cancelled projects and break if management stops the evaluation.

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int q,d,c,s,count=0,on_time=0,total_q=0;
4     for(int i=1;i<20;i++){
5         scanf("%d",&s);
6         if(s==1)break;
7         scanf("%d",&c);
8         if(c==1)continue;
9         scanf("%d %d",&q,&d);
10        total_q+=q;
11        count++;
12        if(d<=0) on_time++;
13        if(q>=90&&d<=2)printf("excellent\n");
14        else if(q>=75 &&q<=89&&d<=5)printf("good\n");
15        else if(q>=60&&q<=74)printf("average\n");
16        else printf("needs improvement\n");
17    }
18    if(count>0){
19        printf("average quality: %.2f\n", (float)total_q/count);
20        printf("on time delivery: %.2f%%\n", ((float)on_time/count)*100);
21    }
22    return 0;
23 }
```

Input

```
0 0 95 1
0 0 88 4
0 0 65 3
0 1
```

Output

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4/4 answered

Q1 Scenario Based - Smart Waste Recycling Plant System

25 marks

A recycling plant processes 5,000 kg of waste every day.

Waste Categories:

Plastic

Paper

Metal

Electronic Waste

Efficiency Levels:

≥90% Recycled — Excellent

75-89% — Good

50-74% — Average

Below 50% — Poor

(a) Develop a C program using arithmetic operators to calculate recycled waste percentage and remaining waste. Use decision-making statements to determine plant efficiency.

(b) Use looping constructs to process waste data for 30 days. Use continue to skip maintenance days and break if the processing machine fails.

Your Code

```
1 #include<stdio.h>
2 int main(){
3     float total=5000,recycled,per,rem;
4     int i,m,f;
5     //part a
6     printf("enter recycled waste:");
7     scanf("%f",&recycled);
8     per=(recycled/total)*100;
9     rem=total-recycled;
10    printf("percentage:%.2f%%\n remaining:%.2fkg\n",per,rem);
11    if(per>=90)printf("excellent");
12    else if (per>=75)printf("good");
13    else if(per>=50)printf("average");
14    else printf("poor");
15    //part b
16    printf("\n\n processing in 30 days\n");
17    for(i=1;i<=30;i++){
18        printf("\n day %d-maintenance ?1 yes 0 no :",i);
19        scanf("%d",&m);
20        if(m==1)continue;
21        printf("machine failed?1 yes 0 no:");
22        scanf("%d",&f);
23        if(f==1){printf("process stopped"); break;}
24    }
25    printf("enter recycled waste:");
26    scanf("%f",&recycled);
27    per=(recycled/total)*100;
28    printf("efficiency:%.2f%%",per);
29 }
30 return 0;
31 }
```

Input

4500
0
0
4000

Output